

Beijing Summer School

Lecture 1: Deep Learning Master Equations for Continuous Time Krusell-Smith Models

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Goals

- Focus on solving Krusell-Smith (2008), which has become a “test” model.
- This lecture: study different proposals for doing projection using neural networks.
- Focus on “*Global Solutions to Master Equations for Continuous Time Heterogeneous Agent Macroeconomic Models.*” by Gu, Lauriere, Payne, and Merkel.

References: Economic Modeling

- Original uninsurable idiosyncratic risk models: Bewley (1980 & 1983), Huggett (1993), Aiyagari (1994) and Krusell and Smith, (1998)
- Continuous time formulations:
 - Achdou, Han, Lasry, Lions, Moll. “Income and Wealth Distribution in Macroeconomics: A Continuous-Time Approach” (2017), and
 - Gu, Lauriere, Merkel, Payne. “Deep Learning Solutions to Master Equations for Continuous Time Heterogeneous Agent Macroeconomic Models” (2023).
- Mean field games: Gomes et al. (2015), Lacker (2015), Carmona and Delarue (2018)

References: Solving PDEs With Deep Learning

- Train neural network to minimize error in differential equation: (This lecture).
 - Raissi-Perdikaris-Karniadakis (2017). “*Physics informed deep learning (part i): Data-driven solutions of nonlinear partial differential equations.*” (Physics Informed Neural Networks (PINNs))
 - Sirignano-Spiliopoulos (2018). “*Deep Galerkin Method (DGM): A deep learning algorithm for solving partial differential equations.*”
- Simulate the economy and learn policies that minimize errors along average path:
 - Han, Jentzen, and Weinan (2018). “*Solving high-dimensional partial differential equations using deep learning*”

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Ingredient 1: Finite Dimensional “Distribution” Approximation

Ingredient 2,3: Remaining Details

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Aside on Calculating Ito Terms

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Aiyagari (1994) and Krusell & Smith (1998)

- Baseline models in “quantitative” macroeconomics.
- Aiyagari-Bewley-Huggett (ABH) (1994,1980,1993): model of precautionary saving:
 - Households get idiosyncratic shocks to labour hours (i.e. employment $\leftrightarrow$ unemployment).
 - Frictions: (i) idiosyncratic labour risk is uninsurable, (ii) exogenous borrowing limit.
 - Households want to borrow when labour hours are low so they can smooth consumption but their borrowing limits prevent it.
 - So, instead, households save assets to sell when labour hours are low
- Krusell & Smith (1998): introduces aggregate TFP shocks into ABH model.
- We are going to setup and solve the continuous time version.

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Setting, Agents and Preferences

- Continuous time, infinite horizon economy with one consumption good.
- Population: continuum of agents, denoted by $I = [0, 1]$.
- Preferences: each agent, $i \in I$, ranks outcomes according to:

$$\mathbb{E}_0 \left[\int_0^\infty e^{-\rho t} u(c_t^i) dt \right],$$

- c_t^i is agent i 's consumption at time, t , and
 - $u(c) = c^{1-\gamma}/(1-\gamma)$ is the flow utility from consuming c
- Household labour endowment, $n_t \in \{n_1, n_2\}$, follows continuous time Markov chain:
 - Process jumps from n_1 to n_2 according Poisson process with intensity rate λ_1
 - Process jumps from n_2 to n_1 according Poisson process with intensity rate λ_2

Production Technology

- Representative firm production technology:

$$Y_t = e^{Z_t} F(K_t, L_t)$$

- K_t, L_t : capital and labor hired at time t
- F : CRS production function (E.g. $F(K, L) = K_t^\alpha L_t^{1-\alpha}$)
- Aiyagari (1994): $Z_t = Z$ is constant so there is no aggregate risk.
- Krussel-Smith (1998): Z_t : aggregate productivity follows a Ornstein-Uhlenbeck process:

$$dZ_t = \eta(\bar{Z} - Z_t)dt + \sigma dB_t^0$$

B_t^0 is a common Brownian motion process; it generates filtration $\mathcal{F}_t^0$.

- Consumption goods can be converted into capital at 1 – 1 ratio (and vice-versa).
- Capital depreciates at rate δ .

Markets

- Incomplete financial markets:
 - Agents cannot insure idiosyncratic income risk.
 - Instead, agents can “save and borrow claims on capital”
 - Let a_t denote agent's claim on capital stock.
 - Face exogenous borrowing constraint: $a_t \geq \underline{a} > -\infty$
- Competitive capital market with rental rate r_t . Clearing implies: $\int_I a_t^i di = K_t$
- Competitive labour market with wage rate w_t . Clearing implies: $\int_I n_t^i di = L_t$
- Competitive goods market (clears by Walras's Law).

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Firm Problem

- Representative firm solves:

$$\max_{K_t, L_t} \{e^{Z_t} F(K_t, L_t) - (r_t + \delta)K_t - w_t L_t\}$$

- First order conditions:

$$(K_t) : \quad r_t = e^{Z_t} \partial_K F(K_t, L_t) - \delta,$$

$$(L_t) : \quad w_t = e^{Z_t} \partial_L F(K_t, L_t)$$

Household Problem

- Household i has **idiosyncratic state** $x_t^i = (a_t^i, n_t^i)$, where a_t^i is wealth, n_t^i is labor.
- Given belief about price process, household chooses **consumption** $c = \{c_t^i\}_{t \geq 0}$ to solve:

$$\begin{aligned} \max_{\{c_t^i\}_{t \geq 0}} \mathbb{E}_0 \left[\int_0^\infty e^{-\rho t} u(c_t^i) dt \right] \\ \text{s.t.} \quad da_t^i = (w_t n_t^i + r_t a_t^i - c_t^i) dt =: s(c_t^i, a_t^i, n_t^i, w_t, r_t) =: \mu_t^a dt, \quad a_t^i \geq \underline{a} \\ n_t^i \in \{n_1, n_2\}, \text{ switches at idiosyncratic Poisson rate } \lambda(l_t^i) \end{aligned} \quad (1)$$

- Households believe prices $q_t := [r_t, w_t]$ follow process $dq_t = \tilde{\mu}_{q,t} dt + \tilde{\sigma}_{q,t} dB_t^0$
- $\underline{a}$: borrowing limit.
- $G_t = \mathcal{L}(x_t^i | \mathcal{F}_t^0)$ & g_t are **population distribution** & **density** of $(x_t^i)_{i \in I}$, given **history** $\mathcal{F}_t^0$
 - Non degenerate because households get uninsurable idiosyncratic labor shocks.

Kolmogorov Forward Equation

- Given processes $\{c, r, w\}$, at time $t > 0$, the density $g_t(a, n)$ satisfies the KFE:

$$dg_t(a, n) = \left[-\partial_a((w_t n + r_t a - c_t)g_t) + \underbrace{\lambda(\check{n})g_t(a, \check{n})}_{\text{Entry from } \check{n} \rightarrow n} - \underbrace{\lambda(n)g_t(a, n)}_{\text{Exit from } n \rightarrow \check{n}} \right] dt, \quad a \in (\underline{a}, \infty)$$

- Questions and observations:
 - Why does the KFE not have dB_t^0 ?
 - What changes to the environment would lead to a dB_t^0 appearing?
 - $g_t(a, n)$ is still a stochastic process in equilibrium since r and w will be stochastic processes adapted to history of common shocks $\mathcal{F}_t^0 = \sigma(\{W_s^0 : 0 \leq s \leq t\})$.

Rational Expectations Equilibrium

Definition: Given initial density g_0 , an **equilibrium** for this economy consists of a collection of $\mathcal{F}_t^0$ -adapted stochastic process, $\{c_t^i, g_t, z_t, q_t := [r_t, w_t] : t \geq 0, i \in I\}$, s.t.:

1. Given belief about price process $(\tilde{\mu}_q, \tilde{\sigma}_q)$, household consumption, c_t^i , solves (1),
2. Given belief about price process $(\tilde{\mu}_q, \tilde{\sigma}_q)$ firm choose capital and labor optimally:

$$r_t = e^{z_t} \partial_K F(K_t, L) - \delta, \quad w_t = e^{z_t} \partial_L F(K_t, L)$$

3. The price vector $q_t = [r_t, w_t]$ satisfies **market clearing conditions** for all t (implying a realized price process $dq_t = \mu_{q,t}dt + \sigma_{q,t}dB_t^0$):

$$K_t = \sum_{j \in \{1,2\}} \int a g_t(a, n_j) da, \quad L = \sum_{j \in \{1,2\}} \int n_j g_t(a, n_j) da$$

4. Agent beliefs about the price process are **consistent**: $(\tilde{\mu}_q, \tilde{\sigma}_q) = (\mu_q, \sigma_q)$

Rational Expectations Equilibrium (Combining Equations For Prices)

Definition: Given an initial density g_0 , an **equilibrium** for this economy consists of a collection of $\mathcal{F}_t^0$ -adapted stochastic process, $\{c_t^i, g_t, q_t := [r_t, w_t], z_t : t \geq 0, i \in I\}$, s.t.:

1. Given belief about price process $(\tilde{\mu}_q, \tilde{\sigma}_q)$, household consumption, c_t^i , solves (1),
2. The price vector $q_t = [r_t, w_t]$ satisfies:

$$q_t = \begin{bmatrix} r_t \\ w_t \end{bmatrix} = \begin{bmatrix} e^{z_t} \partial_K F(K_t, L) - \delta \\ e^{z_t} \partial_L F(K_t, L) \end{bmatrix} =: Q(z_t, g_t), \text{ where } K_t = \sum_{j \in \{1,2\}} \int a g_t(a, n_j) da$$

3. Agent beliefs about the price process are **consistent**: $(\tilde{\mu}_q, \tilde{\sigma}_q) = (\mu_q, \sigma_q)$

Having explicit expression for prices in terms of (z_t, g_t) makes problem very tractable.
(We consider more complicated cases later.)

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Partially Recursive Representation of Equilibrium

- Aggregate states: t , individual states: (a, n) , stoch. household value fn: $V_t(a, n)$.
- Given belief $(\tilde{\mu}_q, \tilde{\sigma}_q)$, household (a, n) chooses c to solve stoch. HJBE:

$$dV_t(a, n) = \max_c \left\{ -\rho V_t(a, n) + u(c) + \partial_a V_t(a, n)(w_t n + r_t a - c) \right. \\ \left. + \lambda(n) (V_t(a, \check{n}) - V_t(a, n)) \right\} dt + \Theta_t(a, n) dW_t^0, \quad s.t. \quad \text{BC: } \frac{\partial V}{\partial a} \Big|_{a=\underline{a}} \geq u'(w_t n + r_t \underline{a})$$

where $\Theta_t(a, n; (\tilde{\mu}_q, \tilde{\sigma}_q))$ is unknown and $\check{n}$ is complement of n .

- For optimal policy, $c_t^*(a, n; (\tilde{\mu}_q, \tilde{\sigma}_q))$, for z_t , population density, g , evolves by KFE:

$$dg_t(a, n) = \underbrace{\left[-\partial_a [(w_t n + r_t a - c_t^*) g_t(a, n)] - \lambda(n) g_t(a, n) + \lambda(\check{n}) g_t(a, \check{n}) \right]}_{=: \mu_t^g(c^*, a, n; (\tilde{\mu}_q, \tilde{\sigma}_q))} dt$$

- Agent beliefs about price process are consistent with clearing: $(\tilde{\mu}_q, \tilde{\sigma}_q) = (\mu_q, \sigma_q)$.

Recursive Representation of Equilibrium

- Aggregate states: (z, g) , individual states: (a, n) , household value fn: $V(a, n, z, g)$.
- Given belief $dg_t = \tilde{\mu}^g(z, g)dt$, household at state (a, n) chooses c to solve **HJBE**:

$$0 = \max_c \left\{ -\rho V(a, n, z, g) + u(c) + \partial_a V(a, n, z, g)(w(z, g)n + r(z, g)a - c) \right. \\ \left. + \lambda(n) (V(a, \check{n}, z, g) - V(a, n, z, g)) + \partial_z V(a, n, z, g)\mu^z(z) + 0.5 (\sigma^z)^2 \partial_{zz} V(a, n, z, g) \right. \\ \left. + \sum_{j \leq 2} \int_{\underline{a}}^{\infty} \frac{\partial V}{\partial g}(a, n, z, g(y, n_j)) \tilde{\mu}^g(y, n_j, z, g) dy \right\}, \quad \text{s.t.} \quad \text{BC: } \frac{\partial V}{\partial a} \big|_{a=\underline{a}} \geq u'(wn + r\underline{a})$$

where $\frac{\partial V}{\partial g}(a, n, z, g(\cdot))$ is Frechet derivative w.r.t. g and $\check{n}$ is complement of n .

- For optimal policy, $c^*(a, n, z, g; \tilde{\mu}^g)$, for z_t , population density, g , evolves by **KFE**:

$$dg_t(a, n) = \underbrace{[-\partial_a [(w(z, g)n + r(z, g)a - c^*)g_t(a, n)] - \lambda(n)g_t(a, n) + \lambda(\check{n})g_t(a, \check{n})]dt}_{=:\mu^g(a, n, z, g; \tilde{\mu}^g)}$$

- In equilibrium, beliefs are consistent $\tilde{\mu}^g = \mu^g$.

Technical Aside: Hamilton-Jacobi-Bellman Equation

- The HJBE is:

$$\begin{aligned} 0 = \max_c \bigg\{ & -\rho V(a, n, z, g) + u(c) + \partial_a V(a, n, z, g)(w(z, g)n + r(z, g)a - c) \\ & + \lambda(n) (V(a, \check{n}, z, g) - V(a, n, z, g)) + \partial_z V(a, n, z, g)\mu^z(z) + 0.5 (\sigma^z)^2 \partial_{zz} V(a, n, z, g) \\ & + \sum_{j \leq 2} \int_{\underline{a}}^{\infty} \frac{\partial V}{\partial g}(a, n, z, g(y, n_j)) \tilde{\mu}^g(y, n_j, z, g) dy \bigg\}, \quad s.t. \quad \text{BC: } \frac{\partial V}{\partial a} \big|_{a=\underline{a}} \geq u'(wn + r\underline{a}) \end{aligned}$$

- Recall technical aside: No “classical solution” because the value function is not differentiable when the borrowing constraint binds. Instead, need to show that a “viscosity” solution exists.
- Recall results (see [Achdou et. al. \(2018\)](#)):
 - Borrowing constraint never binds in the interior of state space
 - Instead, borrowing constraint appears as a boundary inequality

Technical Aside: Kolmogorov Forward Equation

- On $(\underline{a}, \infty)$, the density $g_t(a, y)$ satisfies the KFE:

$$dg_t(a, n) = [-\partial_a[(w(z, g)n + r(z, g)a - c^*)g_t(a, n)] - \underbrace{\lambda(n)g_t(a, n)}_{\text{Exit from } n \rightarrow \check{n}} + \underbrace{\lambda(\check{n})g_t(a, \check{n})}_{\text{Entry from } \check{n} \rightarrow n}]dt$$

- Technical asides:
 - Low income agents get “stuck” at borrowing constraint $\Rightarrow$ mass point in the density.
 - So KFE will not have “classical solution”. Instead, need to use a type of measure value solution (see [Achdou et. al. \(2018\)](#)).
 - The KFE not have dB_t^0 . Why?
 - What changes to the environment would lead to a dB_t^0 appearing in KFE?

Inequality boundary condition causes technical problems $\Rightarrow$ easier to soften it!

Recursive Representation of Equilibrium (Soft Borrowing Constraint)

- Aggregate states: (z, g) , individual states: (a, n) , household value fn: $V(a, n, z, g)$.
- Given a belief $dg_t = \tilde{\mu}_g(z_t, g_t)dt$, household at $x = (a, n)$ choose c to solve **HJBE**:

$$0 = \max_c \left\{ -\rho V(a, n, z, g) + u(c) - \mathbf{1}_{a_t \leq \underline{a}} \psi(a_t) + \partial_a V(a, n, z, g)(w(z, g)n + r(z, g)a - c) \right. \\ \left. + \lambda(n) (V(a, \check{n}, z, g) - V(a, n, z, g)) + \partial_z V(a, n, z, g) \mu^z(z) + 0.5 (\sigma^z)^2 \partial_{zz} V(a, n, z, g) \right. \\ \left. + \sum_{j \leq 2} \int_{\underline{a}}^{\infty} \frac{\partial V}{\partial g}(a, n, z, g) \tilde{\mu}^g(\cdot) dy \right\}, \quad s.t. \quad \text{BC: } \frac{\partial V}{\partial a} \Big|_{a=\underline{a}} \geq u'(\cdot), \quad \psi(a) = -\frac{1}{2} \kappa(a - \underline{a})^2$$

where $\check{n}$ is complement of n .

- For optimal policy, $c^*(a, n, z, g; \tilde{\mu}^g)$, for z_t , population density, g , evolves by **KFE**:

$$dg_t(a, n) = \underbrace{[-\partial_a[(w(z, g)n + r(z, g)a - c^*)g_t(a, n)] - \lambda(n)g_t(a, n) + \lambda(\check{n})g_t(a, \check{n})]}_{=: \mu_t^g(a, n, z, g; \tilde{\mu}^g)} dt$$

- In equilibrium, beliefs are consistent $\tilde{\mu}^g = \mu^g$.

“Master Equation” Representation of Equilibrium

- “Master equation” substitutes KFE, market clearing, & belief consistency into HJBE.
- Equilibrium value function $V(a, n, z, g)$ characterized by one PDE:

$$\begin{aligned} 0 = & -\rho V(a, n, z, g) + u(c^*(a, n, z, g)) + \mathbf{1}_{a_t \leq \underline{a}} \psi(a_t) \\ & + \partial_a V(a, n, z, g)(w(z, g)n + r(z, g)a - c^*(a, n, z, g)) \\ & + \lambda(x) (V(a, \check{n}, z, g) - V(a, n, z, g)) + \partial_z V(a, n, z, g) \mu^z(z) + 0.5 (\sigma^z)^2 \partial_{zz} V(a, n, z, g) \\ & + \sum_{j \leq 2} \int_{\underline{a}}^{\infty} \frac{\partial V}{\partial g}(a, n_j, z, g(y, n_j)) \mu^g(c^*(y, n_j, z, g), y, n_j, z, g) dy =: \mathcal{L}V \end{aligned}$$

where the optimal control c^* is characterised by:

$$u'(c^*(a, n, z, g)) = \partial_a V(a, n, z, g).$$

We find it easier to work with $\partial_a V(a, n, z, g)$ rather than $V(a, n, z, g)$. Why?

Master Equation for $\partial_a V(a, n, z, g)$

- Define $W(a, n, z, g) := \partial_a V(a, n, z, g)$.
- “Master equation” for $W(a, n, z, g)$: (using the envelope theorem.)

$$\begin{aligned}
 0 = & (\textcolor{red}{r}(z, g) - \rho)W(a, n, z, g) + \mathbf{1}_{a_t \leq \underline{a}} \psi(a_t) \\
 & + \underbrace{\partial_a W(a, n, z, g)(\textcolor{red}{w}(z, g)n + \textcolor{red}{r}(z, g)a - \textcolor{blue}{c}^*(a, n, z, g)) + \lambda(l) (V(a, \check{n}, z, g) - W(a, n, z, g))}_{=:\mathcal{L}_x W(a, n, z, g)} \\
 & + \underbrace{\partial_z W(a, n, z, g)\mu^z(z) + 0.5 (\sigma^z)^2 \partial_{zz} W(a, n, z, g)}_{=:\mathcal{L}_z W(a, n, z, g)} \\
 & + \underbrace{\sum_{j \leq 2} \int_{\underline{a}}^{\infty} \frac{\partial W}{\partial g}(a, n_j, z, g(y, n_j)) \mu^g(\textcolor{blue}{c}^*(y, n_j, z, g), y, n_j, z, g) dy}_{=:\mathcal{L}_g W(a, n, z, g)} =: \mathcal{L}W
 \end{aligned}$$

where the optimal control c^* is characterised by: $u'(c^*(a, n, z, g)) = W(a, n, z, g)$.

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Three Notions of equilibrium

- Long running debate about relaxing the rational expectations equilibrium.
- See [Sargent, 2008] and [Sargent, 1999]

Three notions of equilibrium

1. Rational expectations equilibrium
2. Self confirming belief equilibrium (chapter 7 in [Sargent, 1999])
3. Optimal misspecified belief equilibrium (chapter 6 in [Sargent, 1999])

1. Rational expectations equilibrium

- Agents have a “perceived law of motion” for the economic variables that is the true law of motion for the economy if agents follow their perceived law of motion (on and off the equilibrium path).
- In this sense, agents understand the law of motion even for data that would never be generated on the equilibrium path.

Rational Expectations Equilibrium in KS-98

Definition: Given initial density g_0 , an **equilibrium** for this economy consists of a collection of $\mathcal{F}_t^0$ -adapted stochastic process, $\{c_t^i, g_t, z_t, q_t := [r_t, w_t] : t \geq 0, i \in I\}$, s.t.:

1. Given belief about price process $(\tilde{\mu}_q, \tilde{\sigma}_q)$, household consumption, c_t^i , solves (1),
2. Given belief about price process $(\tilde{\mu}_q, \tilde{\sigma}_q)$ firm choose capital and labor optimally:

$$r_t = e^{z_t} \partial_K F(K_t, L) - \delta, \quad w_t = e^{z_t} \partial_L F(K_t, L)$$

3. The price vector $q_t = [r_t, w_t]$ satisfies **market clearing conditions** for all t (implying a realized price process $dq_t = \mu_{q,t}dt + \sigma_{q,t}dB_t^0$):

$$K_t = \sum_{j \in \{1,2\}} \int a g_t(a, n_j) da, \quad L = \sum_{j \in \{1,2\}} \int n_j g_t(a, n_j) da$$

4. Agent beliefs about the price process are **consistent**: $(\tilde{\mu}_q, \tilde{\sigma}_q) = (\mu_q, \sigma_q)$

2. Self confirming belief equilibrium

- Agents have a “perceived law of motion” for the equilibrium variables in the economy.
- In equilibrium, this law of motion is the best statistical fit to data generated by an economy where agents follow that perceived law of motion.
 - So, agents would never reject a misspecification test on their perceived law of motion.
 - However, they may have off-equilibrium beliefs that differ from rational expectations.
- So, this is like an “indoctrination” equilibrium: *agents all believe the world follows particular rules, which influences their behavior and prevents the economy from generating data that could lead them to reject their beliefs.*

2. Self Confirming Belief Equilibrium in KS-98

Definition: Given initial density g_0 , an **equilibrium** for this economy consists of a collection of $\mathcal{F}_t^0$ -adapted stochastic process, $\{c_t^i, g_t, z_t, q_t := [r_t, w_t] : t \geq 0, i \in I\}$, s.t.:

1. Given belief about price process $(\tilde{\mu}_q, \tilde{\sigma}_q)$, household consumption, c_t^i , solves (1),
2. Given belief about price process $(\tilde{\mu}_q, \tilde{\sigma}_q)$ firm choose capital and labor optimally:

$$r_t = e^{z_t} \partial_K F(K_t, L) - \delta, \quad w_t = e^{z_t} \partial_L F(K_t, L)$$

3. The price vector $q_t = [r_t, w_t]$ satisfies **market clearing conditions** for all t (implying a realized price process $dq_t = \mu_{q,t} dt + \sigma_{q,t} dB_t^0$):

$$K_t = \sum_{j \in \{1,2\}} \int a g_t(a, n_j) da, \quad L = \sum_{j \in \{1,2\}} \int n_j g_t(a, n_j) da$$

4. *Agent beliefs about the price process $(\tilde{\mu}_q, \tilde{\sigma}_q)$ are the **best fit** based in any data generated from the model under agent belief $(\tilde{\mu}_q, \tilde{\sigma}_q)$.*

3. Optimal Misspecified Belief Equilibrium: “Dogmatic Form”

- Agents have a “perceived law of motion” for the equilibrium variables in the economy that *satisfies a particular parametric form*.
- In equilibrium, the parameters in their perceived law of motion are the statistical best fit to the data generated by an economy where all agents follow that perceived law of motion.
- However, if agents were to perform a misspecification test on the parametric form for their perceived law of motion, they would generically reject it.

3. Optimal Misspecified Belief Equilibrium in KS-98 (Dogmatic)

Definition: Given initial density g_0 , an **equilibrium** for this economy consists of a collection of $\mathcal{F}_t^0$ -adapted stochastic process, $\{c_t^i, g_t, z_t, q_t := [r_t, w_t] : t \geq 0, i \in I\}$, s.t.:

1. Given belief about price process $(\tilde{\mu}_q, \tilde{\sigma}_q)$, household consumption, c_t^i , solves (1),
2. Given belief about price process $(\tilde{\mu}_q, \tilde{\sigma}_q)$ firm choose capital and labor optimally:

$$r_t = e^{z_t} \partial_K F(K_t, L) - \delta, \quad w_t = e^{z_t} \partial_L F(K_t, L)$$

3. The price vector $q_t = [r_t, w_t]$ satisfies **market clearing conditions** for all t (implying a realized price process $dq_t = \mu_{q,t}dt + \sigma_{q,t}dB_t^0$):

$$K_t = \sum_{j \in \{1,2\}} \int a g_t(a, n_j) da, \quad L = \sum_{j \in \{1,2\}} \int n_j g_t(a, n_j) da$$

4. *Agent beliefs about the price process (i) follow a parametric rule $\check{\mu}_q = \check{\mu}(K_t; \theta)$ and $\check{\sigma}_q = \check{\sigma}(K_t; \theta)$ and (ii) the parameters θ are the “best response” to data generated by the economy under $\check{\mu}_q = \check{\mu}(K_t; \theta)$ and $\check{\sigma}_q = \check{\sigma}(K_t; \theta)$.*

You have seen many deep learning techniques throughout the week.

Which technique do you think correspond to which equilibrium concept?

What the strengths and weaknesses of using each equilibrium concept for evaluating the impact of government policies?

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Numerical Approximation Outline

- Goal: characterize approximate solution to Master equation numerically.
- Problem: Master equation contains an infinite dimensional derivative.
- Solution: three main ingredients:
 1. High but finite dimensional approximation to distribution and Master equation:
 - (i). Replace continuum of agents by a **finite population** of agents, or
 - (ii). **Discretize** the wealth variable, or
 - (ii). **Project** distribution onto a finite dimensional set of basis functions $b_i(x)$ (e.g. eigenfunctions, Chebyshev polynomials, ...).
 2. Parameterize W by neural network, and
 3. Train the parameters to minimize the (approximate) master equation residual.

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Finite Dimensional “Distribution” Approximation

	Finite Population	Discrete State	Project on $\{b_i\}_{i \leq N}$
Dist. approx. params $\hat{\varphi}_t$	Agent states $\hat{\varphi}_t = \{(a_t^i, n_t^i)\}_{i \leq N}$	Masses on grid $\{\hat{\varphi}_{i,t}\}_{i \leq N}$	Basis coefficients $\{\hat{\varphi}_{i,t}\}_{i \leq N}$
Dist. approx. $\hat{g}_t$	$\frac{1}{N} \sum_{i=1}^N \delta_{\hat{\varphi}_t^i}$	$\sum_{i=1}^N \hat{\varphi}_{i,t} \delta_{(a_i, n_i)}$	$b_0(a, n) + \sum_{i=1}^N \hat{\varphi}_{i,t} b_i(a, n)$
KFE approx. ($\mu_{\hat{\varphi}}$)	Evolution of other agents' states	Evolution of mass between grid points (e.g. finite diff.)	Evolution of projec- tion coefficients (least squares)
Dimension (N)	$\approx 20 - 40$	≈ 200	≈ 5
$K_t =$	$\frac{1}{N} \sum_i^N a_t^i$	$\sum_j \sum_{i=1}^N a^i \hat{\varphi}_{i,t}$	$\sum_j \int a \sum_i \hat{\varphi}_{i,t} b_i(a; n_j) da$

Comparison of Approximation Methods

- Dimensionality:
 - **Finite Population**: sufficiently large to capture necessary features of distribution shape
 - **Discrete State**: sufficiently fine to approximate derivatives in KFE well
 - **Projection**: potentially low for efficient basis choice
- Required customization:
 - **Finite Population**: choose number of agents (N)
 - **Discrete State**: choose grid and method for approximating derivatives in KFE on grid
 - **Projection**: choose basis and which aspects of KFE to approximate [see later slides]

Projection is potentially lower-dimensional but requires more intricate choices

1a: Finite Population: Master Equation

$$0 = \mathcal{L}\hat{W}((a^i, n^i), z, \hat{\varphi}) = (r(z, \hat{\varphi}^{-i}) - \rho)\hat{W}((a^i, n^i), z, \hat{\varphi}) + \mathbf{1}_{a \leq a_{ib}} \partial_a \psi(a^i) \\ + (\mathcal{L}_x + \mathcal{L}_z + \hat{\mathcal{L}}_g)\hat{W}((a^i, n^i), z, \hat{\varphi})$$

where the operators become: (using notation $\hat{\varphi}^{-i}$ is $\hat{\varphi}$ excluding element i):

$$\mathcal{L}_x \hat{W}((a^i, n^i), z, \hat{\varphi}) = s((a^i, n^i), \hat{c}^*((a^i, n^i), z, \hat{\varphi}), r(z, \hat{\varphi}^{-i}), w(z, \hat{\varphi}^{-i})) \partial_{a^i} \hat{W}((a^i, n^i), z, \hat{\varphi}) \\ + \lambda(n^i) \left(\hat{W}((a^i, \check{n}^i), z, \hat{\varphi}) - \hat{W}((a^i, n^i), z, \hat{\varphi}) \right)$$

$$\mathcal{L}_z \hat{W}((a^i, n^i), z, \hat{\varphi}) = \partial_z \hat{W}((a^i, n^i), z, \hat{\varphi}) \eta(\bar{z} - z) + \frac{1}{2} \sigma^2 \partial_{zz} \hat{W}((a^i, n^i), z, \hat{\varphi})$$

$$\hat{\mathcal{L}}_g \hat{W}((a^i, n^i), z, \hat{\varphi}) = \sum_{j \neq i} s((a^j, n^j), \hat{c}^*((a^j, n^j), z, \hat{\varphi}), r(z, \hat{\varphi}^{-j}), w(z, \hat{\varphi}^{-j})) \partial_{a^j} \hat{W}((a^i, n^i), z, \hat{\varphi}) \\ + \lambda(n^j) \left(\hat{W}((a^i, n^i), \{(a^j, \check{n}^j), z, \hat{\varphi}^{-j}\}) - \hat{W}((a^i, n^i), z, \hat{\varphi}) \right)$$

Ingredient 1a: Finite Population: Why Are 20-40 Agents Sufficient?

Many finite-agent methods need more agents. Why not here?

1. We sample individual states (x) and distribution states ($\hat{\varphi}$) separately:
 - Without separation: need large N to have sufficiently many x in high-curvature regions (e.g., close to borrowing constraint) to learn value function well.
 - With separation: can independently control where we learn V in the x -dimension and how we represent g .
 - Ultimately, 20-40 agents is sufficient for approximating K because LLN kicks in quickly.
2. When simulating the trained model, we approximate the KFE operator directly instead of simulating finite population economy.

Ingredient 1a: Finite Population: Simulation

- 1: Start from an initial g_0 equal to the ergodic distribution.
 - 2: **for** $j = 0, \dots, N_T - 1$ **do**
 - 3: Sample ΔB_t^0 from normal distribution $N(0, \Delta t)$, construct TFP shock path by:
 $z_{t+\Delta t} = z_t + \eta(\bar{z} - z_t) + \sigma \Delta B_t^0$.
 - 4: **for** $k = 1, \dots, N_{sim}$ **do**
 - 5: Draw states for N agents $\{\varphi_i^k : i = 1, \dots, N\}$ from density g_t at $t = j\Delta t$.
 - 6: At state (z_t, φ_t^k) , get eqm return $r_t^k = \hat{r}(z_t, \varphi_t^k)$ and wage $w_t^k = \hat{w}(z_t, \varphi_t^k)$.
 - 7: **end for**
 - 8: Calculate the average $\bar{r}_t = \frac{1}{N_{sim}} \sum_{k=1}^{N_{sim}} r_t^k$ and $\bar{w}_t = \frac{1}{N_{sim}} \sum_{k=1}^{N_{sim}} w_t^k$
 - 9: At each grid point $a_m \in \underline{a}$, calculate consumption $\hat{c}((a_m, n), z_t, \bar{r}_t, \bar{w}_t), \forall l$.
 - 10: Construct the transition matrix $\mathcal{A}_t$ using finite difference on the grid $\underline{a}$
 - 11: Update g_t by implicit method: $g_{t+\Delta t} = (I - \bar{\mathcal{A}}_t^\top \Delta t)^{-1} g_t$
 - 12: **end for**
-

Ingredient 1b: Discrete State Space: Master Equation

$$0 = \mathcal{L}\hat{W}((a, n), z, \hat{\varphi}) = (r(z, \hat{\varphi}) - \rho)\hat{W}((a, n), z, \hat{\varphi}) + \mathbf{1}_{a \leq a_{lb}} \partial_a \psi(a) \quad (2)$$

$$+ (\mathcal{L}_x + \mathcal{L}_z + \hat{\mathcal{L}}_g)\hat{W}((a, l), z, \hat{\varphi}) \quad (3)$$

where the operators become:

$$\mathcal{L}_x \hat{W}((a, n), z, \hat{\varphi}) = s((a, n), \hat{c}^*((a, n), z, \hat{\varphi}), r(z, \hat{\varphi}), w(z, \hat{\varphi})) \partial_a \hat{W}((a, n), z, \hat{\varphi}) \quad (4)$$

$$+ \lambda(n) \left(\hat{W}((a, \check{n}), z, \hat{\varphi}) - \hat{W}((a, n), z, \hat{\varphi}) \right) \quad (5)$$

$$\mathcal{L}_z \hat{W}((a, n), z, \hat{\varphi}) = \partial_z \hat{W}((a, n), z, \hat{\varphi}) \eta(\bar{z} - z) + \frac{1}{2} \sigma^2 \partial_{zz} \hat{W}((a, n), z, \hat{\varphi}) \quad (6)$$

$$\hat{\mathcal{L}}_g \hat{W}((a, n), z, \hat{\varphi}) = \sum_{j=1,2} \sum_{m=1}^N \mu_{\hat{\varphi}, m, j}(z, \hat{\varphi}) \partial_{\hat{\varphi}_{m, j}} \hat{W}((a, n), z, \hat{\varphi}) \quad (7)$$

where $\partial_{\hat{\varphi}_{m, j}} \hat{W}$ denotes the partial derivative of $\hat{W}$ with respect to the coordinate (m, j) of $\hat{\varphi}$ and where $\mu_{\hat{\varphi}, m, j}(z, \hat{\varphi})$ is the finite difference approximation.

Ingredient 1c: Additional Details on Projection Method

Projection method requires two steps:

1. Finding the law of motion of for the coefficients $\hat{\varphi}_t$

- Goal: find drifts $\mu_{\hat{\varphi},i,t}$, $i = 1, \dots, N$ such that for all $x = (a, l)$

$$\mu_g\left(a, n, z_t, \underbrace{b_0 + \sum_{i=1}^N \hat{\varphi}_{i,t} b_i}_{=\hat{g}_t}\right) \approx \underbrace{\sum_{i=1}^N \mu_{\hat{\varphi},i,t} b_i(a, n)}_{=\partial_t \hat{g}_t(a, l)}.$$

- Problem: there are “infinitely” many equations, but we have only N degrees of freedom.
- We choose $M \geq N$ “test functions” $\phi_1, \dots, \phi_M$ and define the coefficient drifts $\mu_{\hat{\varphi},n}(z, \hat{\varphi})$ as the regression coefficients that minimize the M linear regression residuals

$$\varepsilon_m(z, \hat{\varphi}) := \mu_g\left(a, n, z_t, \underbrace{b_0 + \sum_{i=1}^N \hat{\varphi}_{i,t} b_i}_{=\hat{g}_t}\right) - \sum_{n=1}^N \mu_{\hat{\varphi},i,t} \sum_n \int_A \phi_m(a, n) b_n(a, n) da, \quad m = 1, \dots, M$$

(One “informative” statistic for KS is $\phi(a, n) = a$ to approximate dK_t well)

Ingredient 1c: Additional Details on Projection Method

Projection method requires two steps:

1. Finding the law of motion of for the coefficients $\hat{\varphi}_t$
2. Choosing the basis functions: GLMP-25 chooses:
 - Eigenfunctions of steady-state KFE operator $\bar{\mathcal{L}}^{KF}$ corresponding to largest eigenvalues.
 - If KFE was $dg_t = \bar{\mathcal{L}}^{KF} g_t dt$, these would be the most persistent components of g_t .
 - Actual KFE is $dg_t = \mathcal{L}_t^{KF} g_t dt$; they are still (very) persistent components if $\mathcal{L}_t^{KF} \approx \bar{\mathcal{L}}^{KF}$.
 - ... But there are many other possibilities.

Ingredient 1c: Projection: Master Equation

$$0 = \mathcal{L}\hat{W}((a, n), z, \hat{\varphi}) = (r(z, \hat{\varphi}) - \rho)\hat{W}((a, n), z, \hat{\varphi}) + \mathbf{1}_{a \leq a_{lb}} \partial_a \psi(a) \quad (8)$$

$$+ (\mathcal{L}_x + \mathcal{L}_z + \hat{\mathcal{L}}_g)\hat{W}((a, n), z, \hat{\varphi}) \quad (9)$$

where the operators become:

$$\mathcal{L}_x \hat{W}((a, n), z, \hat{\varphi}) = s((a, n), \hat{c}^*((a, n), z, \hat{\varphi}), r(z, \hat{\varphi}), w(z, \hat{\varphi})) \partial_a \hat{W}((a, n), z, \hat{\varphi}) \quad (10)$$

$$+ \lambda(n) \left(\hat{W}((a, \check{n}), z, \hat{\varphi}) - \hat{W}((a, l), z, \hat{\varphi}) \right) \quad (11)$$

$$\mathcal{L}_z \hat{W}((a, n), z, \hat{\varphi}) = \partial_z \hat{W}((a, n), z, \hat{\varphi}) \eta(\bar{z} - z) + \frac{1}{2} \sigma^2 \partial_{zz} \hat{W}((a, n), z, \hat{\varphi}) \quad (12)$$

$$\hat{\mathcal{L}}_g \hat{W}((a, n), z, \hat{\varphi}) = \sum_{j=1}^N \mu_{\hat{\varphi}, n}(z, \hat{\varphi}) \partial_{\hat{\varphi}_n} \hat{W}((a, n), z, \hat{\varphi}) \quad (13)$$

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Ingredient 2: Approximate V by Neural Network

- Let $\omega = (a, l, z, \hat{\phi})$. We approximate value $V(\omega)$ by neural network with form:

$$\mathbf{h}^{(1)} = \phi^{(1)}(W^{(1)}\omega + \mathbf{b}^{(1)}) \quad \dots \text{Hidden layer 1}$$

$$\mathbf{h}^{(2)} = \phi^{(2)}(W^{(2)}\mathbf{h}^{(1)} + \mathbf{b}^{(2)}) \quad \dots \text{Hidden layer 2}$$

$\vdots$

$$\mathbf{h}^{(H)} = \phi^{(H)}(W^{(H)}\mathbf{h}^{(H-1)} + \mathbf{b}^{(H)}) \quad \dots \text{Hidden layer H}$$

$$V = \sigma(W^{(H+1)}\mathbf{h}^{(H)} + \mathbf{b}^{(H+1)}) \quad \dots \text{Output}$$

- Terminology: a fully connected feed forward NN (finite agent params in blue):
 - H : is the number of *hidden layers*, ($H = 5$)
 - Length of vector $\mathbf{h}^{(i)}$: number of *neurons* in hidden layer i , ($Length = 64$)
 - $\phi^{(i)}$: is the *activation function* for hidden layer i , ($\phi^i = \tanh$)
 - σ : is the *activation function* for the final layer, (soft-plus)
 - $\Theta = (W^1, \dots, W^{(H+1)}, b^{(1)}, \dots, b^{(H+1)})$ are the *parameters*,
- Discrete state, projection more complicated NN [Sirignano and Spiliopoulos, 2018].

Ingredient 3: Algorithm (“EMINN”)

- 1: Initialize the neural networks approximation for $\hat{V}$ with parameters Θ .
- 2: **while** Loss > tolerance **do**
- 3: Sample M new training points: $\mathcal{S} = (a_m, n_m, z_m, \hat{\varphi}_m)_{m=1}^M$.
- 4: Calculate the **weighted average error** across sample points, given current Θ :

$$\mathcal{E}(\mathcal{S}; \Theta) = \kappa^e \frac{1}{M} \sum_{m \leq M} |\hat{\mathcal{L}}(a_m, n_m, z_m, \hat{\varphi}_m; \Theta)|^2 + \kappa^s \mathcal{E}^s(\mathcal{S}; \Theta), \quad \text{where}$$

- $\hat{\mathcal{L}}(a_m, n_m, z_m, \hat{\varphi}_m; \Theta)$ is **error in Master equation** $\hat{\mathcal{L}}$ at m -th training point (derivates in $\hat{\mathcal{L}}$ are calculated using automatic differentiation.)
 - $\mathcal{E}^s(\mathcal{S}; \Theta)$ is **penalty for “wrong” shape** (e.g. penalty for non-concavity of V)
- 5: Update parameters Θ by **stochastic gradient descent**: $\Theta^{new} = \Theta^{old} - \alpha D_{\Theta} \mathcal{E}(\mathcal{S}; \Theta)$
 - 6: **end while**
-

The Macroeconomic Deep Learning Conundrum

- Algorithm is straightforward to describe and code
- ...but not necessarily easy to implement successfully!
- I will discuss some features that we have found helpful.

Sampling Approaches

- Sampling (a, n, z) : draw from uniform; add draws where error high (active sampling).
- Sampling the parameters in the distribution approximation $(\hat{\varphi}_i)_{i \leq N}$:
 - *Moment sampling*:
 1. Draw samples for selected moments of the distribution (that are important for $Q(z, \hat{g})$).
 2. Sample $\hat{\varphi}$ from a distribution that satisfies the moments drawn in the first step.
 - *Mixed steady state sampling*:
 1. Solve for the steady state for a collection of fixed aggregate states z .
 2. Draw random mixtures of this collection of steady state distributions.
 - *Ergodic sampling*:
 1. Simulate economy using current value function approximation.
 2. Use simulated distributions as training points.

Need to choose economically relevant subspace on which to sample.

Other Implementation Aspects

- Bad approximate solutions (“cheat solutions”)
 - Neural network training may get stuck in bad approximate solutions.
(e.g. constant, non-increasing, or non-concave V)
 - We deal with this by **penalizing shape** that corresponds to known bad solutions.
 - Alternative: train ϕ satisfying $V(a, n, z, g) = \phi(a, n, z, g; \theta)(a - \underline{a})^{1-\gamma}$ instead of V
 - We also explore complementary ways of addressing this.
(initialization, sampling, network architecture)
- Boundary conditions
 - Previous algorithm assumes no boundary condition.
 - To apply it, we **“soften”** hard constraints in the model.
 - Alternative: draw sample S^b on boundary and add boundary error $\mathcal{E}^b$ to loss function.
But: hard constraints represent significant difficulty for robustness of our algorithm.

Some Implementation Details

	Finite Population	Discrete State	Projection
Neural Network			
(i) Structure	Feed-forward	Recurrent + embedding	Recurrent + embedding
(ii) Initialization	$W(a, \cdot) = e^{-a}$	random	random
Sampling			
(i) (a, n)	Active sampling $[\underline{a}, \bar{a}] \times \{n_1, n_2\}$	Uniform sampling $[\underline{a}, \bar{a}] \times \{n_1, n_2\}$	Uniform sampling $[\underline{a}, \bar{a}] \times \{n_1, n_2\}$
(ii) $(\hat{\varphi}_i)_{i \leq N}$	Moment sampling: sample r , then random agents positions that give r	Mixed steady-state sampling then ergodic sampling	Sample K and then orthogonal coefficients from ergodic sampling
(iii) z	$U[z_{min}, z_{max}]$	$U[z_{min}, z_{max}]$	$U[z_{min}, z_{max}]$
Loss Function			
(i) Constraints	$\partial_{aa} V(a, \cdot), \partial_{za} V(a, \cdot) < 0$	$\partial_{aa} V(a, \cdot), \partial_{za} V(a, \cdot) < 0$	$\partial_{aa} V(a, \cdot), \partial_{za} V(a, \cdot) < 0$
(ii) Weights	$\kappa^e = 100, \kappa^s = 1$	$\kappa^e = \kappa^s = 1$	$\kappa^e = \kappa^s = 1$

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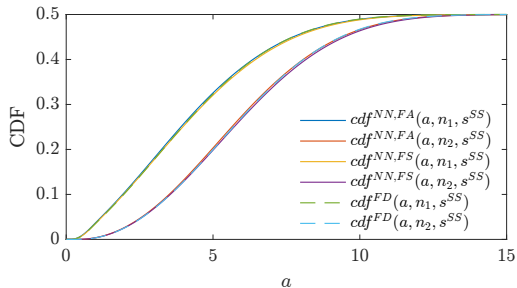
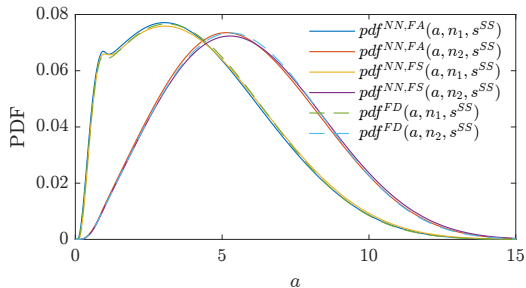
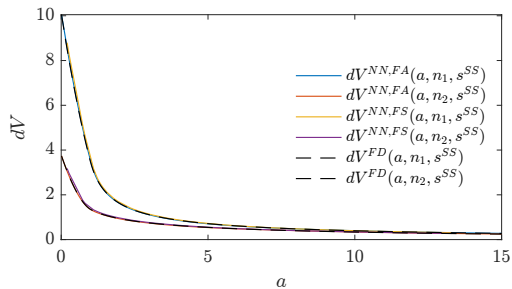
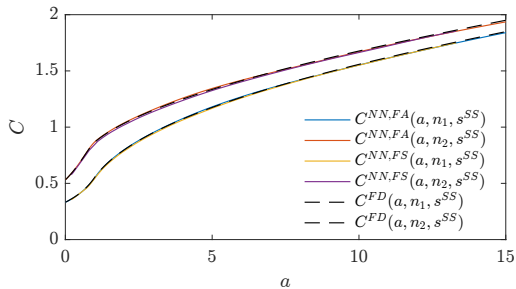
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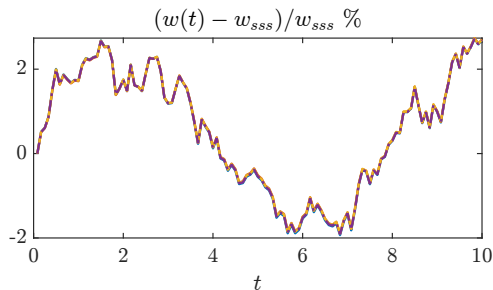
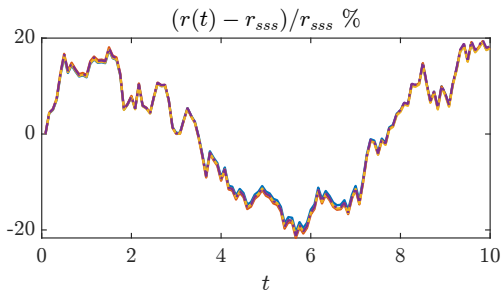
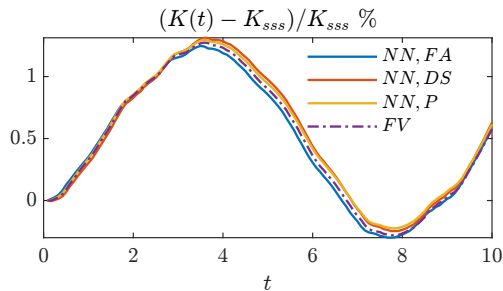
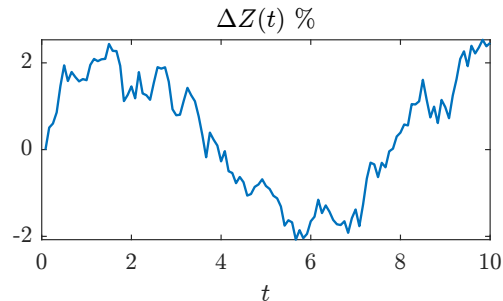
Spatial Business Cycle Model (Time Permitting)

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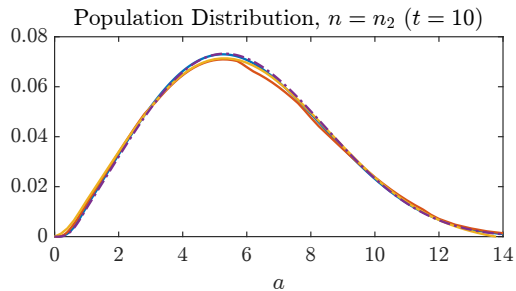
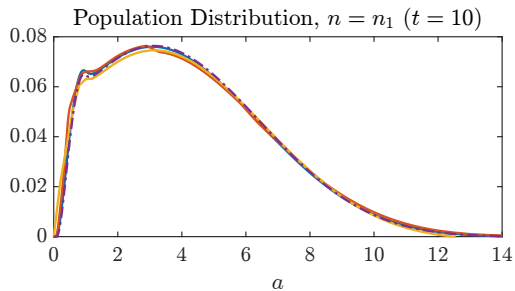
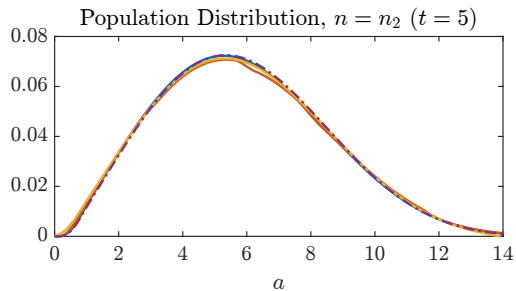
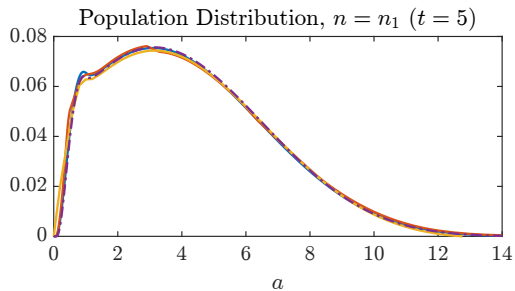
Aiyagari Model: Neural Network and Finite Difference Solution



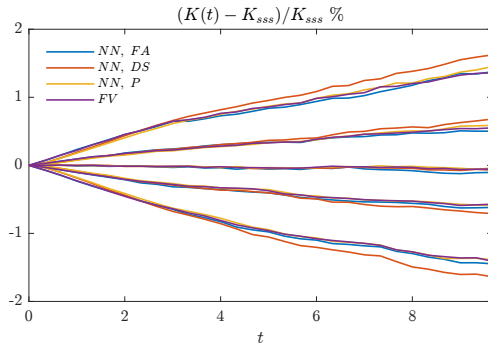
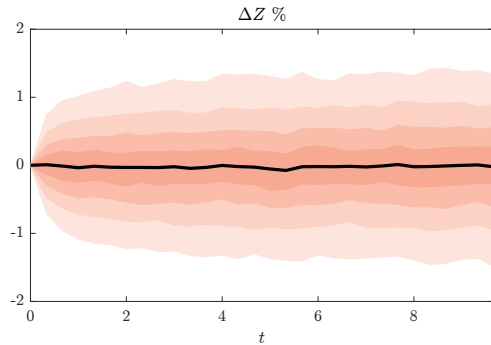
Krusell-Smith: Sample Time Paths



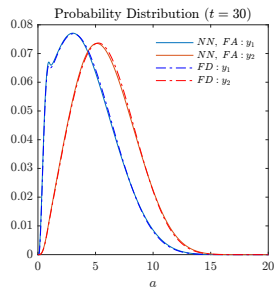
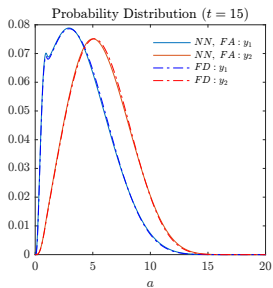
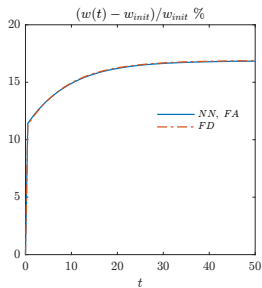
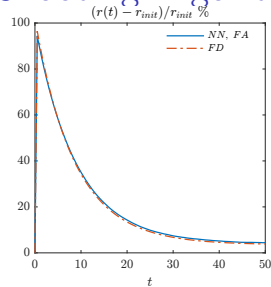
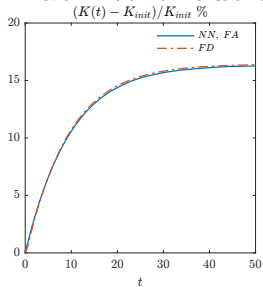
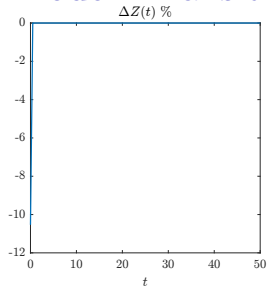
Krusell-Smith: Population Distribution



Krusell-Smith: Fan Chart



Aiyagari Model: Transition Path Without a Shooting Algorithm



Reflections on Different Techniques

- Finite agent was robust in a number of ways:
 - The neural network can be trained with the moment sampling procedure
 - The algorithm only requires a moderate number of agents (approximately 40).
- Discrete-state method to be difficult to work with:
 - Having to approximate the derivatives in the KFE is hard.
 - One challenge is that this requires a fine grid for agent wealth.
 - Also, the training samples need to come from relatively smooth densities and so ergodic sampling is very important. This made the approach slow and complicated.
- Projection method brings a different set of trade-offs
 - Other approaches feed relatively large state spaces into the neural network and then let the neural network work out how to structure the approximate value function.
 - By contrast, the projection method requires more choices ex-ante.

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Ingredient 1: Finite Dimensional “Distribution” Approximation

Ingredient 2,3: Remaining Details

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Conclusion

Concerns About The Computation of “Ito-terms”

- Complaint: EMINN methods require too much differentiation.
- Reason: consider the more abstract version of the master equation:
 - Let the state space be (x, s) where $x := (a, n)$ and $s := (z, \hat{\varphi})$ collects the aggregate states.
 - Let the $\mu(s)$ and $\sigma(s)$ denote the drift and volatility of s .
 - Then, in general form, the master equation involves:

$$\begin{aligned} 0 = & (r(s) - \rho)\hat{W}(x, s) + \mathbf{1}_{a \leq a_{lb}} \partial_a \psi(a^i) \\ & + \mu_a(x, s) \partial_a \hat{W}(x, s) + \lambda(n)(\hat{W}((a, \check{l}), s) - \hat{W}((a, n), s)) \\ & + D_s \hat{W}(x, s)^T \mu(s) + 0.5 \text{Tr}[\sigma(s)^T D_s^2 \hat{W}(x, s) \sigma(s)] \end{aligned}$$

- This potentially involves computing a large number of first and second order derivatives if φ is high dimensional.

Option 1

- Take a low dimensional projection
- E.g. Choose test function that approximates the law of motion for the first moment of g well and then take 2 other eigenvectors.
- E.g. Project onto the first and second moments of g .
- E.g. Project onto the first principle component.

Option 2 (Potentially Combined With Option 1)

- [Duarte, 2018] points out there is a convenient way to compute Ito terms.
- For a given s , define the auxiliary function $F : \mathbb{R} \rightarrow \mathbb{R}$ by:

$$F(\epsilon) \equiv \hat{W} \left(s + \frac{\epsilon}{\sqrt{2}} \sigma(s) + \frac{\epsilon^2}{2m} \mu(s) \right)$$

Then:

$$D_s \hat{W}(x, s)^T \mu(s) + 0.5 \text{Tr}[\sigma(s)^T D_s^2 \hat{W}(x, s) \sigma(s)] = F''(0)$$

- So, we can evaluate the Ito terms with limited differentiation.

Additional Models That Fit The GLMP-25 Framework

1. Heterogeneous firm dynamics:

- In KS-98, there is only one firm.
- We extend our method to models with many firms that each have non-tradable capital and capital adjustment costs (a version of [Khan and Thomas, 2008])
- Technical difference is that we need to price firm equity.

2. Dynamic spatial model:

- In the KS-98 model, there is one wage in the entire economy
- We extend our method to a spatial model where there are different wages in different locations and agents can move (a version of [Bilal, 2021]).
- Technical difference is that the wage in location j only depends on the part of the population distribution at location j .

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Environment: High Level

- In KS-98, there is one wage and rental rate for the entire economy.
- Now study a spatial model with separate wage/rental markets and agent movement.
- Bilal (2024) with investment, capital mobility, and solved globally without linearizing.
- Tech. difference: wage, rental rate in location j only depend on part of distribution.

Environment: Setting

- Continuous time with infinite horizon. Finite set of locations $j \in \{1, \dots, J\}$.
- Continuum of workers $i \in [0, 1]$ and continuum of capital owners $i \in [1, 2]$.
- Representative firm at each location j produces consumption goods according to:

$$Y_{j,t} = \exp(\beta_j + \chi_j z_t) K_{j,t}^{\alpha_k} L_{j,t}^{\alpha_l}, \quad \alpha_k + \alpha_l < 1$$

- z_t is aggregate TFP, which follows $dz_t = \eta(\bar{z} - z_t)dt + \sigma dB_t^0$
- $\beta_j \geq 0$ is a time-invariant location-specific productivity shifter,
- $\chi_j \geq 0$ governs sensitivity of production in location j to aggregate productivity z_t .
- Each location has labor and capital markets with wage $w_{j,t}$ and rental rate $r_{j,t}$:

$$w_{j,t} = w_j(z_t, g_t) := \alpha_l \exp(\beta_j + \chi_j z_t) (g_t^k(j))^{\alpha_k} (g_t^l(j))^{\alpha_l - 1}.$$

$$r_{j,t} = r_j(z_t, g_t) := \alpha_k \exp(\beta_j + \chi_j z_t) (g_t^k(j))^{\alpha_k - 1} (g_t^l(j))^{\alpha_l}.$$

Environment: Workers and Capital Owners

- Each worker $i \in [0, 1]$ has discount rate ρ and utility $u(c_t^i) = (c_t^i)^{1-\gamma}/(1-\gamma)$.
 - Endowed with one unit of labor, which they supply inelastically in their location.
 - Idiosyncratic state $x_t^i = j_t^i$ consists solely of the location of residence.
 - At rate $\mu > 0$, gets moving opportunity and i.i.d. Gumbel location amenity draw.
 - Moving from j to j' also incurs cost $\tau_{j,j'}$.
 - So, j_t^i follows a continuous-time Markov chain with transition rate $\mu\pi_{j,j',t}$ from j to j'

$$\pi_{j,j',t} = \pi_{j,j'}(V_t) := \frac{e^{\nu(V_t(j') - \tau_{j,j'})}}{\sum_{k=1}^J e^{\nu(V_t(k) - \tau_{j,k})}}, \quad V_t(j) \text{ is worker value function at location } j$$

- Capital owners have same location choice problem but also build capital.
 - Capital stock of capital owner $i \in (1, 2]$ evolves according to $dk_t^i = r_{j_t^i,t}k_t^i - c_{j_t^i,t}^i$,
 - Can aggregate cross sectional capital distribution within but not across locations.

Master Equations

- The worker value function $V(j, z, g)$ solves the master equations:

$$0 = -\rho V(j, z, g) + u(w_j(z, g)) + (\mathcal{L}_x + \mathcal{L}_z + \mathcal{L}_g)V(j, z, g), \quad \text{where}$$

$$\mathcal{L}_x V(j, z, g) = \mu \left(\frac{1}{\nu} \log \left(\sum_{j'=1}^J e^{\nu(V(j', z, g) - \tau_{j, j'})} \right) - V(j, z, g) \right)$$

$$\mathcal{L}_z V(j, z, g) = \partial_z V(j, z, g) \eta(\bar{z} - z) + \frac{1}{2} \sigma^2 \partial_{zz} V(j, z, g), \quad \mathcal{L}_g V(j, z, g) = \sum_{j'=1}^J \partial_{g(j')} V(j, z, g) \mu_g(j', z, g),$$

- Capital producer Master equation for $W(j, k, z, g)$ is similar but with optimal consumption choice ζ^* .
- Population density g^l and capital density g^k solve the KFEs:

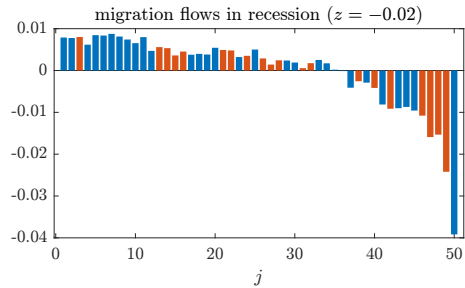
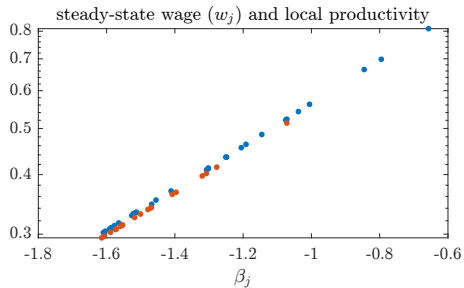
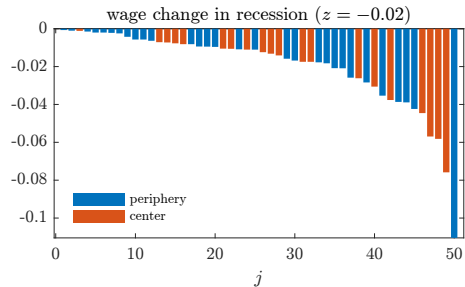
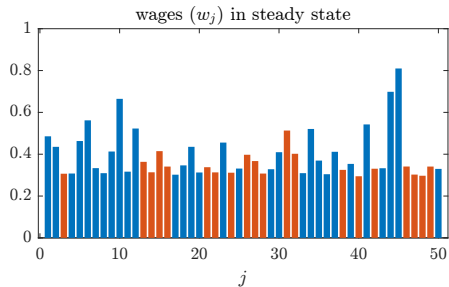
$$\mu_{g^l}(j, z, g) = \mu \left(\sum_{\iota=1}^J \pi_{\iota, j} (V(\cdot, z, g)) g^l(\iota) - g^l(j) \right)$$

$$\mu_{g^k}(j, k, z, g) = \mu \left(\sum_{\iota=1}^J \pi_{\iota, j} (W(\cdot, z, g)) g^k(\iota, k) - g^k(j, k) \right) - \partial_k \left(\left(r(j, z, g) - \zeta^*(j, z, g) \right) k g^k(j, k) \right)$$

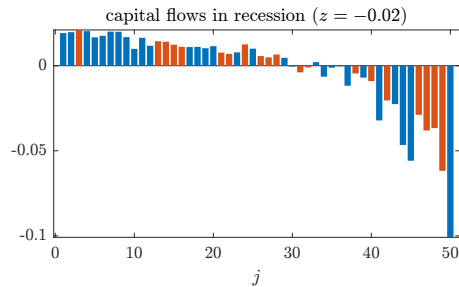
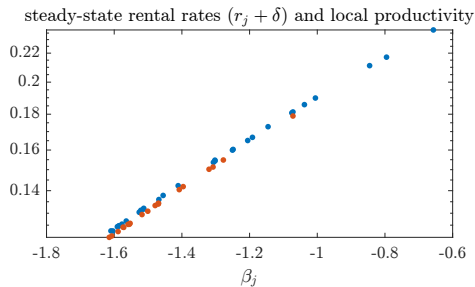
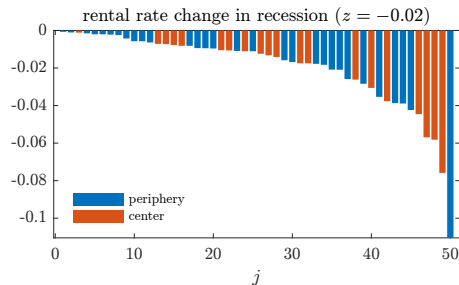
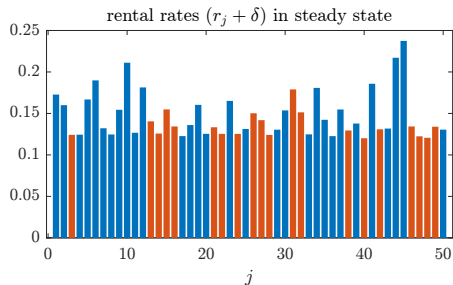
Computation and Calibration

- Approximate $V(j, z, g), W(j, k, z, g)$ by neural networks and train to solve Master eqs.
- We necessarily use the discrete state space density representation because the problem is naturally discrete and shape is very important.
- Calibration:
 - Use use 50 locations.
 - We choose the $\tau_{j,j}$ distribution to have a core-periphery structure.
 - We choose χ_j distribution to match the wage elasticities in Bilal (2024).

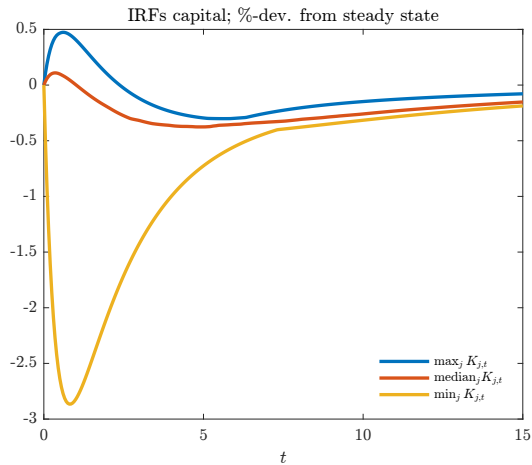
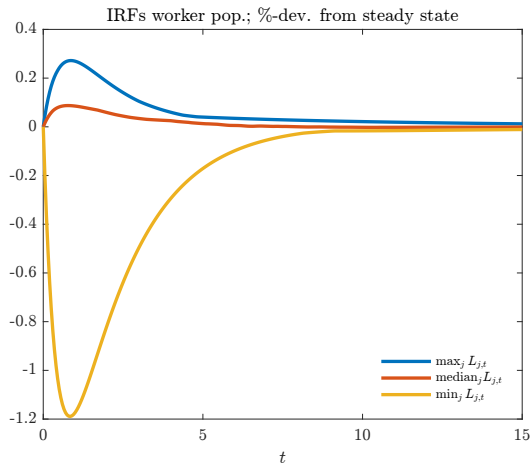
Wages and Population Flows in a Recession (Ranked by χ_j)



Rental Rates and Capital Flows in a Recession (Ranked by χ_j)



Worker and Capital Flows in a Recession



Worker and Capital Flows in a Recession (All Locations)

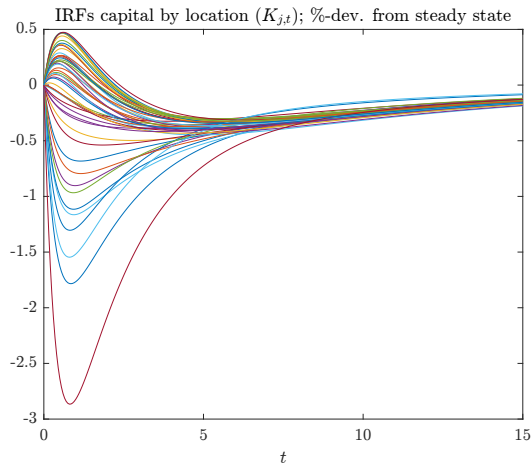
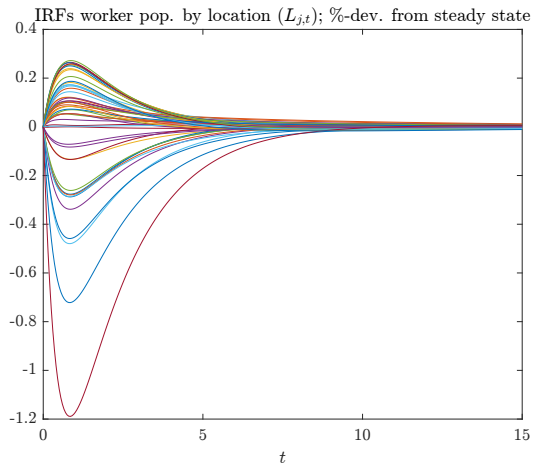


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Summarizing Practical Lessons on Solving Master Equations

- Explained algorithm to characterize rational expectations solutions to heterogeneous agent models with aggregate shocks.
- Three methods for approximating population distribution
 - Which method works best depends on how distribution interacts with agent decision making and complexity of KFE.
 - For KS model: all methods effective, but finite agent approach is simplest and appears most robust.
- Implementation details matter! Some practical lessons:
 1. Working out the correct sampling approach is very important.
 2. Neural networks find it easier to work with smooth constraints.
 3. Enforcing shape constraints helps with speed and stability.
 4. Starting with a simple, solvable model is helpful for tuning hyperparameters.

Conclusion

- Setup the Master equation for KS-98 under rational expectations.
- Discussed different approaches for solving KS-98 with deep learning.
- Main differences are in how to approximate the distribution.

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